

MLFF common online-monitor specification

Status: current normative monitoring policy

Restored P5 scope: exact campaign-common target monitor plus independent replay monitor

1. Scope and semantic type boundary

This specification defines deterministic campaign-common **monitoring evidence**. It does not define target-training size, replay-training membership, checkpoint score thresholds, or held-out cross-validation roles.

The current policy families are type-distinct:

OnlineTargetMonitorPolicy
ReplayMonitorPolicy
ResolvedTargetSizePolicy

Their records SHALL remain distinct even when two cardinalities share an integer value. Online monitors never supply gradients.

2. Common target monitor

OnlineTargetMonitorPolicy owns the current campaign-common target checkpoint monitor used by restored P5. The exact monitor record is constructed once and reused across every selected target size, CV fold, CV seed, and final-production seed/run belonging to the same current campaign/method lineage.

The current P5 values are:

requested cardinality = 256
realized cardinality = 256
seed = 161803
parent role = OUTER_MONITOR

Fewer than 256 usable eligible parent configurations is typed P5 infeasibility. There is no current P5 short-parent success state and no shrink-to-fit fallback.

The exact deterministic stratification, quota, marker, ordering, and systematic-selection semantics are D2 authority. The record SHALL preserve enough evidence to reconstruct the exact ordered membership.

Parent role and label usability

The target-monitor parent is the neutral authorized label-usable OUTER_MONITOR domain. Gradient-training, held-out CV evaluation, calibration, purge-only, excluded, and locked-test roles are not eligible parents.

The monitor constructor validates label-domain compatibility and target metric label usability before a current record is admitted.

Membership before relation qualification

Monitor membership and target/monitor protected-relation separation are distinct responsibilities.

The required order is:

1. select exact 256 membership from the label-usable neutral parent under the accepted D2 monitor algorithm;
2. persist/authenticate the common monitor record; then
3. use canonical P1 protected-relation authority over the joint monitor/target universe to prove separation from every governed T_N.

A relation conflict causes failure. The constructor/planner SHALL NOT relation-prefilter the parent, delete/replace a conflicting selected member, or resample another monitor.

SelectedRelationProjection alone is insufficient to prove this cross-role separation because it intentionally excludes frames outside T_{selected} .

3. OnlineTargetMonitorRecord

The current record binds at least:

target-monitor role
neutral parent identity/digest
OnlineTargetMonitorPolicy digest
requested size = 256
realized size = 256
exact ordered selected identities/source indices
accepted D2 stratum/quota/order evidence
strategy/seed identity
label-domain identity
label-usability evidence
exact membership/content digest

A current CV/final plan additionally binds the plan-level protected-relation separation evidence; that separation evidence is not folded into monitor membership identity because it depends on the governed selected target set(s).

Every current sibling CV/final plan SHALL bind the same common-monitor record digest.

4. Independent true-label replay monitor

ReplayMonitorPolicy owns replay-monitor construction. The current default requested replay-monitor cardinality is:

512 configurations

Replay monitoring is independent of the common target monitor and of replay-training membership.

Foundation pseudo-label replay is explicit opt-in. When used for training, an independent TRUE_DFT replay monitor is required for retention evidence.

Replay train and replay monitor configurations remain geometry/role separated under their governing contract. Current replay-monitor fallback behavior, where separately accepted, does not imply any fallback for the P5 target monitor.

5. P5 owner binding

Current restored P5 binds target monitor identity through PostSelectionMethodIdentity descendants and the CV/final plans described in `mlff_post_selection_p5_spec.md`.

Broad historical/general TrainingProtocolIdentity monitor fields do not create a second current P5 owner. Historical monitor records may remain readable but cannot authorize current P5 if they encode fold-local target monitors, different parents, short-parent target success, or other superseded semantics.

6. Leakage and independence invariants

- Monitoring configurations never contribute gradients.
- The common target monitor is development/model-control evidence, not held-out CV evidence.
- Held-out CV evaluation and locked tests cannot be promoted into monitor roles.
- Exact common-monitor membership is reused; it is not redrawn per size, fold, seed, epoch, or final run.
- Protected-relation qualification occurs after exact membership construction and cannot mutate membership.
- The monitor seed/policy is identity, not an informal hint.
- Monitor cardinality is never interpreted as N_{selected} .

- Common-monitor labels may control checkpoints/adaptive stop and contribute representative target metrics, but they do not enter P5 residual-E0 fitting.

7. Relationship to target-size study

P3 target-size screening may have separately accepted development-monitor consumption, but it does not own current restored-P5 monitor construction and cannot turn monitor cardinality into target size.

A P5-only monitor-generation/currentness cutover SHALL NOT blanket-invalidate unchanged P3 evidence unless the actual P3 monitor semantics it consumes changed.

8. Precision and accumulation boundary

Monitor inference uses the learned model precision/backend bound by the applicable current method/runtime identity. mdstats-owned metric accumulation remains under its numerical-precision authority.

Changing a numerically material precision/backend field changes the relevant method/runtime identity where required by current contracts.

9. Persistence and unsupported historical records

Current target-monitor artifacts are accepted only when their current schema, exact content, neutral parent lineage, policy digest, label usability, and exact-256 requirement validate.

Obsolete fold-local, alternate-parent, M3-derived, short-target-monitor, or otherwise incompatible target-monitor records remain historical. Compatibility deserialization cannot make them current.

10. Acceptance requirements

Current monitor qualification covers at least:

1. deterministic exact target membership regeneration;
2. exact target requested/realized size 256 with no short-parent success;
3. common target-monitor identity across all current sibling CV/final plans;
4. accepted D2 stratum/quota/time-selection behavior;
5. label-domain/metric-label usability;
6. post-sampling protected-relation failure without resampling;
7. proof that selected-only relation projection is not the sole cross-role authority;
8. independent TRUE_DFT replay-monitor enforcement where required;
9. current P5 plan/identity binding;
10. corruption/staleness rejection; and
11. proof that monitor cardinalities are not consumed as target-size authority.